



Semester I Examinations 2015/16

Exam Code(s)	1SD1, 1SD3, 1MF1
Exam(s)	Higher Diploma (Software Design & Development) M.Sc. (Software Design & Development) Higher Diploma (Software Design & Development) – Industry Stream
Module Code(s)	CT516 CT861 CT538 CT542
Module(s)	Computer Architecture and Operating Systems
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor Liam Maguire
Internal Examiner(s)	Professor Gerard Lyons Dr. Michael Madden *Dr. Michael Schukat

Instructions: Answer two questions from Section A and one question from Section B.

Duration	2 hrs
No. of Pages	4
Discipline(s)	Information Technology
Requirements	None

Release to Library: Yes ☒ No ☐

Section A

Question 1

- a) What is an *interrupt* and what is meant by the term *interrupt processing*?
4 MARKS
- b) What is the difference between the *sign-magnitude* and the *two's complement* representation of a negative integer value?
Convert the four integer values 1, -13, 27 and -126 into 8-bit values using both representations.
8 MARKS
- c) Using examples distinguish between the process scheduling algorithms
- first come first served
- shortest process next
- shortest remaining time
- round robin.
12 MARKS
- d) Distinguish between the two I/O techniques
- programmed I/O
- interrupt-driven I/O.
6 MARKS

Question 2

- a) Using examples distinguish between the secondary storage management techniques
- contiguous allocation
- chained allocation.
6 MARKS
- b) Explain how information is stored on a magnetic disk consisting of multiple platters.
6 MARKS
- c) Using an example discuss the purpose of a *conditional jump* instruction as part of a CPU instruction set.
8 MARKS
- d) Summarise the *7-state process model* and, using an example, show how a process moves through all different states.
10 MARKS

PTO

Question 3

- a) Using an example summarise the interactions between a CPU, its cache and the main memory. In your answer refer to the internal structure of the cache.
10 MARKS
- b) What is meant by the term *locality of reference*? How is it related to cache design?
4 MARKS
- c) Distinguish between the instruction types:
- data transfer
- arithmetic
- logical
- conversion
- transfer of control
6 MARKS
- d) Summarise the main features of the memory management technique *demand paging*. Using an example distinguish between the two page replacement policies
- First in, first out
- Least recently used
10 MARKS

PTO

Section B

Question 4

- a) Distinguish between the transport layer protocols *TCP* and *UDP*. In your answer discuss content and structure of the headers of both protocols.
10 MARKS
- b) Using an example discuss how the *basic bit-map protocol* implements a collision-free protocol.
8 MARKS
- c) Briefly summarise similarities and differences between a *repeater*, a *hub* and a *switch*.
6 MARKS
- d) Using an example summarise how a firewall operates on packet-level.
6 MARKS

Question 5

- a) Using examples show TCP uses a 3-way handshake to establish and to release a connection.
10 MARKS
- b) *Network address translation* (NAT) is a common feature of modern network routers. Explain how NAT works and how it can protect a network from unsolicited external access.
8 MARKS
- c) Summarise the purpose and inner workings of the
 - address resolution protocol.
 - dynamic host configuration protocol8 MARKS
- d) Distinguish between the concepts of *static* and *dynamic routing* algorithms.
4 MARKS